

```
from collections import deque
```

```
def bfs(graphs,start):
```

```
    visited = set()
```

```
    queue = deque([start])
```

```
    visited.add(start)
```

```
    order=[]
```

```
    while queue:
```

```
        vertex = queue.popleft()
```

```
        order.append(vertex)
```

```
        for neighbor in graphs[vertex]:
```

```
            if neighbor not in visited:
```

```
                visited.add(neighbor)
```

```
                queue.append(neighbor)
```

```
    return order
```

```
def dfs(graphs,start,visited=None,order=None):
```

```
    if visited is None:
```

```
        visited = set()
```

```
    if order is None:
```

```
        order = []
```

```
    visited.add(start)
```

```
    order.append(start)
```

```
for neighbor in graphs[start]:  
    if neighbor not in visited:  
        dfs(graphs, neighbor, visited, order)
```

```
return order
```

```
#example useage
```

```
if __name__ == "__main__":
```

```
    graphs = {  
        'A': ['B', 'C'],  
        'B': ['A','D', 'E'],  
        'C': ['A','F'],  
        'D': ['B'],  
        'E': ['B','F'],  
        'F': ['C','E']  
    }
```

```
print("BFS Traversal starting from 'A':", bfs(graphs, 'A'))
```

```
print("DFS Traversal starting from 'A':", dfs(graphs, 'A'))
```

```
print("\n-----")
```

```
print("Program executed successfully by:")
```

```
print("Name: Bhuvan K")
```

```
print("Roll No: [24BECS118]")
```

```
print("-----")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code + -

```
● PS C:\Users\91638\OneDrive\Documents\Ai> python -u "c:\Users\91638\OneDrive\Documents\Ai\tempCodeRunnerFile.py"
BFS Traversal starting from 'A': ['A', 'B', 'C', 'D', 'E', 'F']
DFS Traversal starting from 'A': ['A', 'B', 'D', 'E', 'F', 'c']

-----
Program executed successfully by:
Name: Bhuvan K
Roll No: [24BECS118]
-----
❏ PS C:\Users\91638\OneDrive\Documents\Ai>
```